ADEDOYIN SAMUEL AJEYOMI

Address: Abuja FCT, Nigeria
Email: adedoyinsamuel25@gmail.com
Google Scholar | Portfolio | LinkedIn | Github

CAREER OBJECTIVES

To build and deploy end-to-end data solutions that transform raw, complex datasets into reliable decision-making tools, covering the full data lifecycle, data collection, engineering, modeling, analytics, visualization, and reporting. My goal is to become a leader in the data and research sector, driving practical, scalable, and impactful solutions for environmental, development, and real-world challenges, rather than focusing solely on algorithm development.

WORK/PROFESSIONAL EXPERIENCE

Data Engineer Consultant (Contract)
International Water Management Institute (IWMI) **March 2026 – August 2026**

- Designed and maintained the canonical data schema, data dictionary, and integration mappings supporting the Demand Intelligence Platform (DIP) and Enabling Environment Digital Toolbox.
- Built repeatable, versioned ETL pipelines to ingest, normalize, validate, and load datasets from policy documents, surveys, workshop reports, and innovation repositories.
- Implemented automated data quality systems including schema validation, duplicate detection, anomaly checks, broken-link monitoring, and validation reporting.
- Developed integration pipelines with CGIAR knowledge systems, including CGSpace and CGIAR Performance and Results Management System (PRMS), enabling interoperable data exchange and innovation–demand linkage.
- Established data operations infrastructure with scheduled ingestion workflows, monitoring, logging, and runbooks to support the transition from PoC to operational MVP platforms used across global CGIAR programs.

Data Engineer/ Dashboard Developer
Spatialnode Inc, Delaware, USA (Project-Based Contract). **August, 2025 – January, 2026**

- Worked on the **Multidimensional Inclusivity Index (MDII)** and **Demand Intelligence Information System (DIIS)** projects, delivering end-to-end data and analytics solutions for decision support.
- Handled the full data lifecycle across both projects, including data ingestion, cleaning, preprocessing, transformation, validation, and storage using DAX, Python, and Azure Functions.
- Built and maintained CI/CD pipelines to automate data processing workflows, dashboard updates, and report refreshes, ensuring reliability and reproducibility.
- Developed interactive Power BI dashboards and analytical reports, enabling evidence-based decision-making across Nigeria, Bangladesh, Ethiopia, and other countries.
- Processed, standardized, and analyzed 10,000+ records from multi-source, unstructured, and semi-structured datasets, improving data quality and usability.
- Extracted and structured critical indicators and insights from documents, and collaborated with stakeholders to define metrics, reporting requirements, and analytical outputs.

Geospatial Data Engineer/Backend Developer (Contract)
Envirotrust (https://envirotrust.eu/), Germany (Remote) **June 2025 to March 2026**

Built scalable ETL pipelines and RESTful APIs to process and serve large geospatial datasets (Zarr, COG, Parquet) using Xarray, DuckDB, FastAPI, and Titiler for real-time visualization and analysis.

- Deployed cloud-native infrastructure on AWS (S3, Lambda, EC2) with Docker and CI/CD (GitHub Actions) to automate, scale, and maintain geospatial workflows.
- Led collaborative development on GitHub, ensuring clean code, agile iteration, and reliable deployment across remote teams.
- Engineered data pipelines to source and aggregate diverse geospatial data for wildfire, flood, heat, and wind risk modeling, preparing it for cloud-based analysis.
- Developed scalable data aggregation strategies for environmental risk parameters, optimizing data transfer to cloud platforms for advanced scenario modeling.

Geospatial Data Analyst / Trainer **2022 – 2025**

Integral Bridge Solutions (Part-time), Abuja, FCT, Nigeria

Responsibilities:

- Led the development of proposals, workplan designs, and technical report writing
- Trained clients and end users in using our in-house solutions and various GIS tools
- Led a team of analysts on Geospatial and Statistical Analysis and Modelling using various data formats, both spatial and non-spatial.
- Performed advanced geospatial analysis and modeling using machine learning techniques and Google Earth Engine.
- Automating traditionally manual geospatial workflows through scripting and process optimization.

Geospatial Analyst (Hybrid), **2023 – 2024**

Milsat Technologies Limited, Abuja, Nigeria

Responsibilities:

- Performed advanced data analytics to uncover insights and trends, enhancing decision-making accuracy.
- Led the Design and management geospatial and statistical dashboards for real-time monitoring and analysis.
- Developed and optimized ETL processes for efficient data extraction, transformation, and loading.
- Implemented and maintained databases to improve data storage and retrieval.
- Led the designed unbiased survey questionnaires to enhance response accuracy and reliability.

Geospatial Research Assistance **July 2020 – September 2020**

The Nigeria Erosion and Watershed Management Project (NEWMAP)

Responsibilities:

- Analyzed water flow patterns using geospatial data to assess flood risks and water resource management.
- Developed flood hazard maps by integrating satellite imagery, elevation models, and hydrological data.
- Created digital elevation models (DEMs) for infrastructure planning, erosion control, and terrain analysis.
- Conducted land use classification and watershed delineation to study environmental changes and water management.
- Assisted teams in learning GIS tools, remote sensing techniques, and data visualization for effective decision-making.

EDUCATION

- **Bachelor of Technology in Remote Sensing and Geoscience Information System** **2016 - 2023**
  Federal University of Technology Akure (FUTA), Nigeria

PROFESSIONAL CERTIFICATE

- **WorldQuant University** – Applied Data Science Lab **2025**
  https://www.credly.com/badges/6fcb79c5-11a5-4eac-bfb4-c723f9d42419
- **AWS Cloud Practitioner** **2023**
  https://www.credly.com/badges/8783fbc3-adeb-4462-9ae2-f1d170a09536/public_url

TECHNICAL SKILLS

Data Engineering & Programming: Python (Pandas, GeoPandas, Gdal, Shapely, Rasterio, DuckDB, Scikit-learn, FastAPI), SQL (data modeling, querying, transformations), R, Bash, Backend development and API-based data services

Data Analytics & Visualization: Excel, Power BI (data modeling, dashboards, reporting), KPI development, analytical workflows, insight generation

Cloud, DevOps & Infrastructure: AWS, GCP, Azure Functions, and cloud-based data pipelines, CI/CD pipelines, workflow automation and Docker

Geospatial & Remote Sensing (Supporting): Google Earth Engine (GEE), QGIS, ESRI Suite (ArcGIS Pro, ArcPy, ArcGIS Online), OpenStreetMap (OSM), JOSM

PROFESSIONAL SKILLS

- Research writing, methodology development, and result interpretation
- Technical documentation, reporting, and presentation

- Stakeholder engagement and requirements gathering
- Communicating complex data insights to technical and non-technical audiences
- Problem-solving, analytical thinking, and collaboration

LANGUAGES

English – Fluent (speaking, reading, writing, listening)**, Yoruba –** Native

HONORS, AWARDS AND RECOGNITIONS

- Recipient, **META/DSN Geospatial Health Mobility Hackathon (Nigeria) ($1000)** [Link].2023
- Recipient, **ESRI Young Scholar (YSA) (Nigeria)** 2023
- Recipient, **Mappa Data Engineering Project Competition (£250)** 2023
- Recipient, **URISA Top 3 position in the 2023 GIS Pro Digital Competition** 2023
- Recipient, **Jamitan/OSGOF GIS Day 2023 ($200USD) and a 1year ESRI License** 2023

PROFESSIONAL MEMBERSHIP

- Professional Member, Spatial and Data Science Society of Nigeria **2019 to Date**

LEADERSHIP AND VOLUNTEERING EXPERIENCE

Training and Capacity Development Lead **2023 to Date**

Spatial and Data Science Society of Nigeria (SDSN)

- Facilitated several educational programs such as workshops, and continuing to creating contents to enlighten people on GeoAI.
- Mentoring and engaging new professionals within the geospatial domain.
- Support the development and upgrading of the society website while managing and coordinating the process.

GIS Analyst **2021 – 2022**

Omdena Lagos Chapter Nigeria [Certificate]

- Worked on local community projects, conducting geospatial analysis of pipe-borne water availability in Lagos and electricity energy distribution in Nigeria
- Provided Geospatial Support on another Project where we try to Estimate and understand Electricity availability and accessibility using GeoML.
- Worked as one of the Public Relations Officer of the Group

OpenStreetMap Contributor **2019 – 2023**

YouthMapper FUTA Member

- Facilitated community building and collaboration while conducting OSM mapping training sessions, successfully recruiting new members to expand impact.
- Contributed to Data Geospatial Digitization towards Humanitarian Purpose through HOTOSM
- Served as the Welfare Coordinate to manage and oversee the welfare of the community members

GIS Analyst | Field Data Collector Officer **2021 - 2022 (8 months**

YouthMapper Research Follow Project (Wastespy)

- Designed and implemented questionnaires using KoboToolbox and mapped illegal dumpsites in Ibadan with KoboCollect, performing spatial analytics for insights.
- Validated, cleaned, and migrated field data to databases using Python.
- Created reports and visualizations to communicate findings to stakeholders.

REFEREES

Available upon Request